Understanding Deep Learning

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Chapter 1: Introduction

Artificial intelligence includes approaches such as search, logic, and probabilistic reasoning. Machine learning fits mathematical models to data, and deep learning fits a particular family of models, deep neural networks. These models can be used in supervised, unsupervised, and reinforcement learning.

1.1 Supervised learning

A supervised model maps an input to a prediction. Real objects must first be encoded numerically: for example, a house becomes a vector of features, a sentence a sequence of word indices, and an image an array of pixel values. The numerical output is interpreted in the context of the task.

- Regression predicts continuous values, such as a house price. Multivariate regression predicts several values, such as a molecule's freezing and boiling points.
- Classification predicts a category. Binary classification has two categories; multiclass classification has more. The model can output a probability for each class.

Inputs differ in size and structure. Tabular features have no intrinsic order, provided their interpretation changes consistently with any rearrangement. Word order matters in text, while images have spatial relationships between pixels, and molecules have variable numbers of atoms and bonds. Audio and images can also be very high-dimensional. A suitable model must accommodate the structure of its input.

A model is a *family* of input-output relationships, indexed by parameters. Training selects a relationship that describes labeled input-output pairs well. For instance, pairs of children's ages and heights can determine a curve used to predict height at a new age. The pairs act as supervision. Deep networks represent broad families of relationships and can be fitted to data; they can handle large, variable-length, or structured inputs and outputs.

1.1.1 Structured outputs

Outputs can themselves have many dependent components. Semantic segmentation assigns a class to each pixel, while depth estimation predicts a depth at each pixel. In both, output structure is closely related to the input image: neighboring similar pixels often have related outputs. Speech transcription, translation, and caption-to-image generation also have structured outputs, but their correspondence with the input is less direct.

These latter tasks have two complications. There may be several valid outputs for one input, and outputs must obey structural constraints: arbitrary word sequences need not form sentences, and arbitrary pixel values need not form plausible images. Learning the output's structure from unlabeled data can help, connecting supervised and unsupervised learning.

1.2 Unsupervised learning

Unsupervised learning uses data without corresponding output labels to describe or understand its structure. The data may be discrete or continuous, high- or low-dimensional, and fixed- or variable-length.

A generative model learns to produce new examples with statistics resembling those of the training data. Some models explicitly specify a probability distribution and sample from it; others provide a generation mechanism without explicitly specifying the distribution. Conditional generation holds some information fixed while generating the rest: image inpainting fills a missing region given the remaining pixels, and text completion continues a given prefix. A plausible completion is not a guarantee that its contents are true.

1.2.1 Latent variables

Observed data can have far more coordinates than the underlying factors needed to describe its variation. Realistic images occupy only a small subset of all possible pixel arrays; physical processes constrain their structure. For example, facial expression varies with the movements of a comparatively small number of muscles.

Some generative models represent an example using underlying latent variables z and a learned mapping g to data x. A simple distribution over z gives a way to generate samples:

$$z \sim p(z), \qquad x = g(z).$$

Latent variables need not correspond to identifiable physical causes. If representations for two examples can be found, interpolating between them and mapping the intermediate points back to data space can produce plausible intermediate examples. Plausibility is a property to assess, not a guarantee for every interpolation.

1.2.2 Connecting the two settings

For a structured prediction task such as caption-to-image generation, one can learn representations of text and images from unlabeled data and then learn a relationship between the representations using paired examples. Lower-dimensional representations may reduce the number of pairs needed, and an image generator can impose useful structure on its outputs. Introducing randomness allows

different plausible images for the same caption. These are potential advantages, not guarantees that every generated image is valid or matches its caption.

1.3 Reinforcement learning

In REINFORCEMENT LEARNING, an agent repeatedly takes actions in an environment. Actions affect the state, possibly stochastically, and produce rewards. The aim is to choose actions that yield high reward over time on average. A deep network can represent a POLICY, mapping an observed state to an action or a distribution over actions.

Two difficulties distinguish this setting:

- TEMPORAL CREDIT ASSIGNMENT: a reward can arrive long after the actions that contributed to it. A robot rewarded at a checkpoint must learn which earlier movements helped; a chess player rewarded only at the end must assign credit across many moves.
- EXPLORATION VERSUS EXPLOITATION: an agent can use a known strategy that earns some reward or try alternatives that might do better. A robot that has learned to slide forward may need to explore balancing and walking to improve.

The reward specification matters: for chess, rewarding captured pieces and rewarding only the final outcome provide different feedback. An opponent's responses can also make the effective transition from a chosen move uncertain.

1.4 Ethics

The benefits and harms of deep learning depend on its data, design, deployment, and control. Historical data can reproduce or amplify unfair treatment; for example, a salary predictor trained on biased past salaries may perpetuate those disparities. Large models are difficult to explain: a local explanation may describe one decision without explaining the entire system.

Other concerns include military use, concentration of power and economic disruption, surveillance, privacy violations, fraud, disinformation, and environmental costs. The chapter also raises the possibility of loss of control over more capable systems as an existential risk; the scale and likelihood of such future harms are uncertain. Researchers and deployers should examine both intended and unintended uses, organizational incentives, and whether institutions have practical mechanisms to stop harmful projects.

Source. Simon J.D. Prince, Understanding Deep Learning, Chapter 1, pp. 1–16 (PDF pp. 15–30), February 2026 version.